Physics 210 Tutorial #7 Multiple Choice Problems on Magnetic Forces and Fields

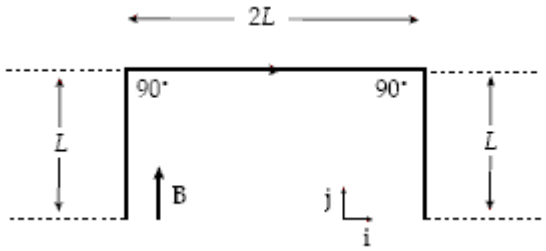
____ 1. A particle (mass = 2.0 mg, charge = –6.0 μC) moves in the positive direction along the *x* axis with a velocity of 3.0 km/s. It enters a magnetic field of (2.0**i** + 3.0**j** + 4.0**k**) mT. What is the acceleration of the particle?

a. (36**j** – 27**k**) m/s^2
b. (–36**j** + 27**k**) m/s^2
c. (–24**j** + 18**k**) m/s^2
d. (24**j** – 18**k**) m/s^2
e. (24**j** – 27**k**) m/s^2

____ 2. A segment of wire carries a current of 25 A along the *x* axis from *x* = –2.0 m to *x* = 0 and then along the *y* axis from *y* = 0 to *y* = 3.0 m. In this region of space, the magnetic field is equal to 40 mT in the positive *z* direction. What is the magnitude of the force on this segment of wire?

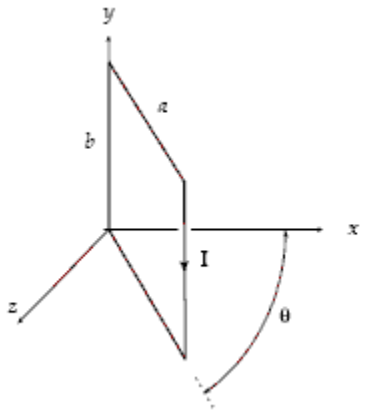
a. 2.0 N
b. 5.0 N
c. 1.0 N
d. 3.6 N
e. 3.0 N

____ 3. A straight wire is bent into the shape shown. Determine the net magnetic force on the wire when the current *I* travels in the direction shown in the magnetic field **B**.



a. 2*IBL* in the –*z* direction
b. 2*IBL* in the +*z* direction
c. 4*IBL* in the +*z* direction
d. 4*IBL* in the –*z* direction
e. zero

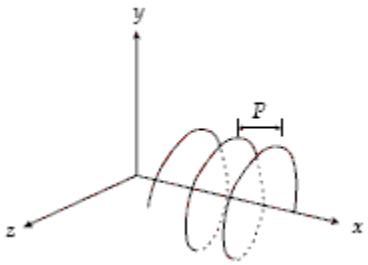
____ 4. The figure shows the orientation of a rectangular loop consisting of 80 closely wrapped turns each carrying a current *I*. The magnetic field in the region is (40 **i**) mT. The loop can turn about the *y* axis. If $\theta = 30°$, $a = 0.40$ m, $b = 0.30$ m, and $I = 8.0$ A, what is the magnitude of the torque exerted on the loop?



a. 2.5 N · m
b. 1.5 N · m
c. 3.1 N · m
d. 2.7 N · m
e. 0.34 N · m

____ 5. What current must be maintained in a square loop (50 cm on a side) to create a torque of 1.0 N · m about an axis through its center and parallel to one of its sides when a magnetic field of magnitude 70 mT is directed at 40° to the plane of the loop?
a. 66 A
b. 89 A
c. 61 A
d. 75 A
e. 37 A

____ 6. A uniform magnetic field of 0.50 T is directed along the positive *x* axis. A proton moving with a speed of 60 km/s enters this field. The helical path followed by the proton shown has a pitch of 5.0 mm. Determine the angle between the magnetic field and the velocity of the proton.



a. 39°
b. 51°
c. 44°
d. 34°
e. 71°

____ 7. A particle (m = 3.0 μg, q = 5.0 μC) moves in a uniform magnetic field given by (60**j**) mT. At t = 0 the velocity of the particle is equal to (30**j** – 40**k**) m/s. The subsequent path of the particle is
   a. circular with a 50-cm radius.
   b. helical with a 6.3-cm pitch.
   c. circular with a period of 31 ms.
   d. helical with a 40-cm radius.
   e. none of the above

____ 8. A proton moves around a circular path (radius = 2.0 mm) in a uniform 0.25-T magnetic field. What total distance does this proton travel during a 1.0-s time interval? ($m = 1.67 \times 10^{-27}$ kg, $q = 1.6 \times 10^{-19}$ C)
   a. 82 km
   b. 59 km
   c. 71 km
   d. 48 km
   e. 7.5 km

____ 9. An electron follows a circular path (radius = 15 cm) in a uniform magnetic field (magnitude = 3.0 G). What is the period of this motion?
   a. 0.12 μs
   b. 1.2 ms
   c. 0.18 μs
   d. 1.8 ms
   e. 1.8 μs

____ 10. A proton is accelerated from rest through a potential difference of 2.5 kV and then moves perpendicularly through a uniform 0.60-T magnetic field. What is the radius of the resulting path?
   a. 15 mm
   b. 12 mm
   c. 18 mm
   d. 24 mm
   e. 8.5 mm

____ 11. What value of B should be used in a velocity selector to separate out 2.0-keV protons if E is fixed at 80 kV/m?
   a. 0.18 T
   b. 0.11 T
   c. 0.15 T
   d. 0.13 T
   e. 0.23 T

Physics 210 Multiple Choice Week 8
Answer Section

MULTIPLE CHOICE

1. ANS: A PTS: 1
2. ANS: D PTS: 1
3. ANS: B PTS: 1
4. ANS: D PTS: 1
5. ANS: D PTS: 1
6. ANS: B PTS: 1
7. ANS: D PTS: 1
8. ANS: D PTS: 1
9. ANS: A PTS: 1
10. ANS: B PTS: 1
11. ANS: D PTS: 1